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#include <stdio.h>
#include "platform.h"
#include "xil_printf.h"
#include "xscugic.h"
#include "xparameters.h"
#include "xscuwdt.h"
#include "xil_exception.h"

XScuWdt_Config *wdt_config;
XScuWdt wdt;

XScuGic_Config *gic_config;
XScuGic gic;

void wdtHandler(){

    if(XScuWdt_IsTimerExpired(&wdt))
    {
        xil_printf("We have an Interrupt at 1 Sec\n");
    }

}

void wdtInit(){
    int status = 0;
    wdt_config = XScuWdt_LookupConfig(XPAR_PS7_SCUWDT_0_DEVICE_ID);
    status = XScuWdt_CfgInitialize(&wdt,wdt_config,wdt_config->BaseAddr);
    if(status == XST_SUCCESS)
        xil_printf("WDT Init Successful\n");
    else
        xil_printf("WDT Init Failed\n");

    XScuWdt_Stop(&wdt);
    XScuWdt_SetTimerMode(&wdt);
    XScuWdt_LoadWdt(&wdt,333333333); //1sec
    u32 opt = XScuWdt_GetControlReg(&wdt);
    opt = opt | XSCUWDT_CONTROL_IT_ENABLE_MASK | XSCUWDT_CONTROL_AUTO_RELOAD_MASK;
    XScuWdt_SetControlReg(&wdt,opt);

}

void gicInit(){
    int status = 0;
    gic_config = XScuGic_LookupConfig(XPAR_SCUGIC_0_DEVICE_ID);
    status = XScuGic_CfgInitialize(&gic, gic_config, gic_config->CpuBaseAddress);
    if(status == XST_SUCCESS)
        xil_printf("GIC Init Successful\n");
    else
        xil_printf("GIC Init Failed\n");

    Xil_ExceptionInit();
    Xil_ExceptionRegisterHandler(XIL_EXCEPTION_ID_IRQ_INT, (Xil_ExceptionHandler) XScuGic_InterruptHandler, &gic);
    Xil_ExceptionEnable();

    XScuGic_Connect(&gic, XPAR_SCUWDT_INTR, (Xil_InterruptHandler)wdtHandler, &wdt);
    XScuGic_Enable(&gic, XPAR_SCUWDT_INTR);

}

int main()
{
    init_platform();
    wdtInit();
    gicInit();
    XScuWdt_Start(&wdt);

    while(1);

    cleanup_platform();
    return 0;
}

```